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Charges Module - Fino

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KARPLEXUS

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Overview

The banks offer a range of financial products and services to their customers, and each service may incur specific charges. The Banking Charges Module is designed to manage and monitor the various charges associated with banking services. The Charges Module helps to streamline the process of calculating, applying, and tracking these charges, ensuring transparency and compliance with regulatory standards.

Key features

1. **Charges Structure Configuration:**

   Allows administrators to define and configure a flexible charges structure for different services such as cash transactions, payment transactions, etc.

2. **Override:**

   Supports override charges strategies based on the customer, product, or other dynamic factors.

3. **Waivers:**

   Provides functionality to apply waivers for specific customers, transactions, or under certain conditions.

4. **Thresholds and Limits:**

   Allows setting thresholds and limits that trigger the application of charges, such as transaction frequency limits.

5. **Effective Date Management:**

   Enables setting effective dates for charges, ensuring that charges are applied only during specific time period.

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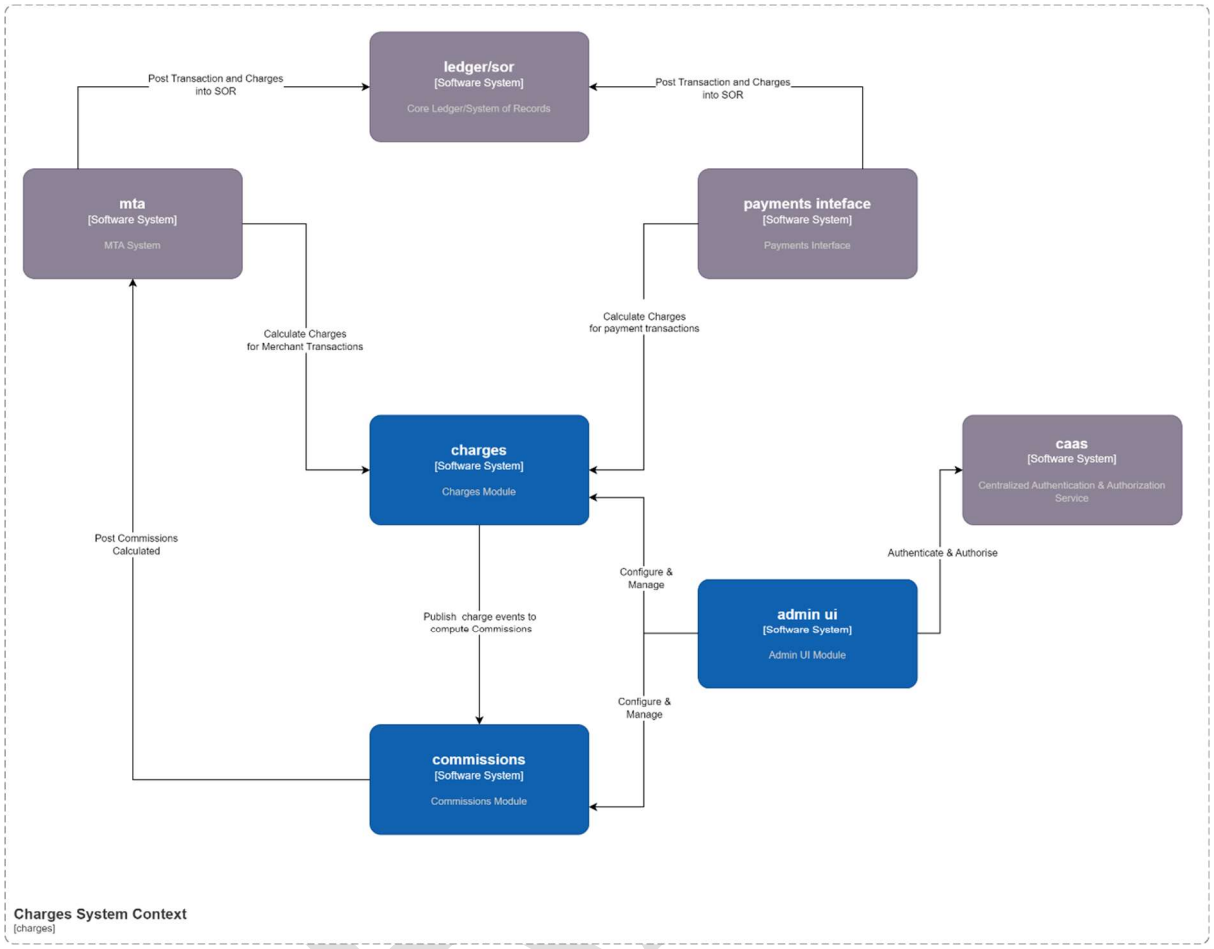
Context

The below diagram shows the overall context of the Charges Module.



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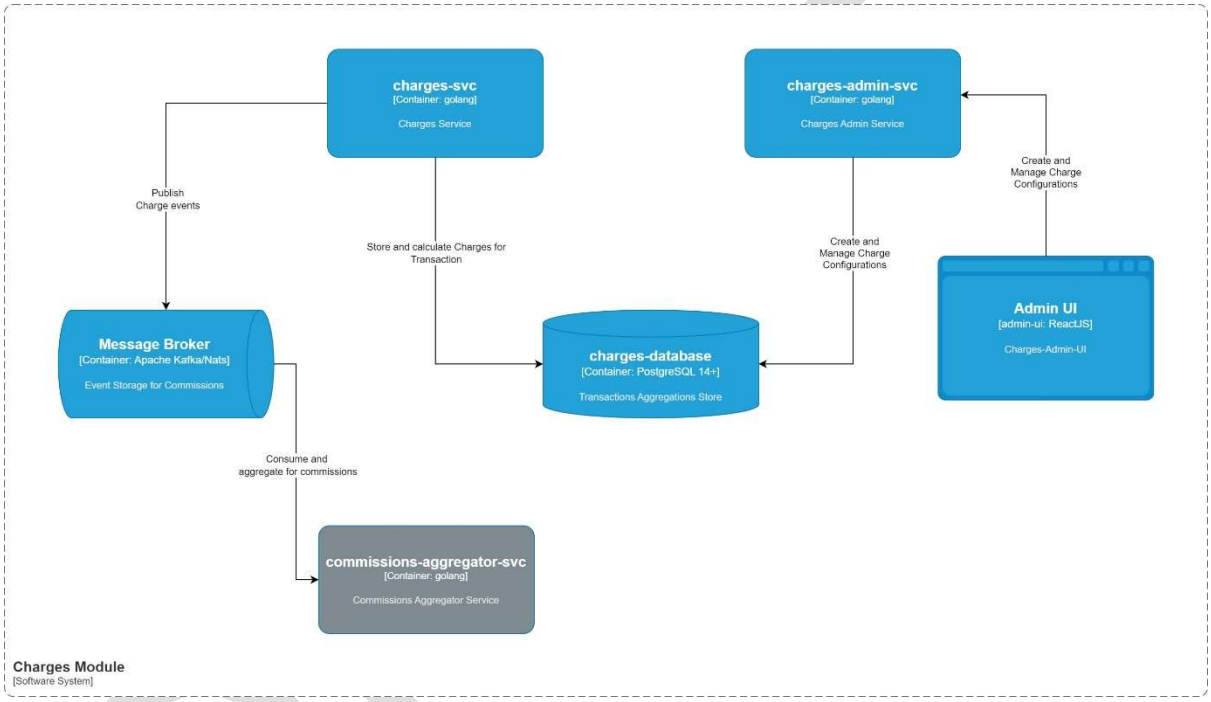
Charges module workflow

Charges module has the following primary components:

Admin UI and Charges-admin-svc - Charge admin service is to create and manage charge configurations.

Charges-svc - Charges-svc stores and calculate charges for the transaction.

Charges-database - To store transaction aggregations.



List of Entities

Following is the list of entities referred to in the document as part of various business processes,

Principle:

The charges module manages and applies charges for various services, configuring charge structures, adhering to regulations, and utilizing dynamic pricing. Supports and provides the override and waives charges based on the customer category, transaction type, product type, or under certain conditions. Automated workflows streamline processes, enhancing overall operational efficiency.

Transaction Type:

Refers to the specific type of financial transaction or activity for which a charge is applied.

Product Type:

The category or type of financial product associated with a charge.

Customer Category:

Classifies customers into different categories based on criteria like account balance, transaction frequency, or relationship status. Charges may vary depending on the customer category.

Charge Plan:

Represents a predefined strategy or structure for applying charges. Charge plans outline how charges are calculated and under what conditions they are applied.

Charge:

The actual amount charged for a specific banking service or transaction.

Flat:

A type of charge structure where the charge is a fixed, unchanging amount regardless of the transaction amount or other factors.

Percentage:

A type of charge structure where the charge is calculated as a percentage of the transaction amount. The charge amount varies proportionally with the transaction value.

Min & Max:

Specifies the minimum and maximum limits for a charge. If a charge is subject to these limits, it ensures that the charge applied is within the defined range.

Overrides:

Refers to instances where predefined charge structures or rules are overridden or modified for specific conditions. Overrides allow for flexibility in handling unique situations.

Override Attributes:

Describes the specific parameters or characteristics that trigger an override in the charge structure. These attributes could include customer status, transaction type, or other conditions that warrant a deviation from the standard charge application.

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The Operational Framework

The operational framework of the charges module demonstrates the dynamic functionality and systematic application of charges.

As an administrator, designing charge codes involves crafting a flexible and responsive framework that aligns with the diverse needs of the financial institution. In this context, key attributes such as transaction type and effective date play a pivotal role, while incorporating flags for override, waivable, and tax introduces a layer of sophistication to the calculation process.

Transaction Type and Effective Date Attributes:

The essence of charge code design begins with understanding the transaction types and effective dates. Transaction types describe the nature of financial activities, whether it be IMPS, RTGS, or other services. Effective dates, on the other hand, introduce temporal considerations, allowing administrators to set specific periods during which charges are applicable.

Flags for Override, Waivable, and Tax:

Override Flag:

The override flag empowers administrators to deviate from standard charge structures based on exceptional circumstances. It provides the flexibility to modify charges for specific transactions or customer segments, ensuring a particular approach to charge application.

Waivable Flag:

The waivable flag offers a customer-centric feature, allowing administrators to designate charges that can be waived under certain conditions. This enhances customer satisfaction by providing the institution with the ability to offer exemptions or discounts based on predefined criteria.

Tax Flag:

The tax flag integrates tax considerations into the charge code design. By flagging charges subject to taxation, the system automates the calculation and application of taxes, streamlining compliance with tax regulations and reducing manual intervention.

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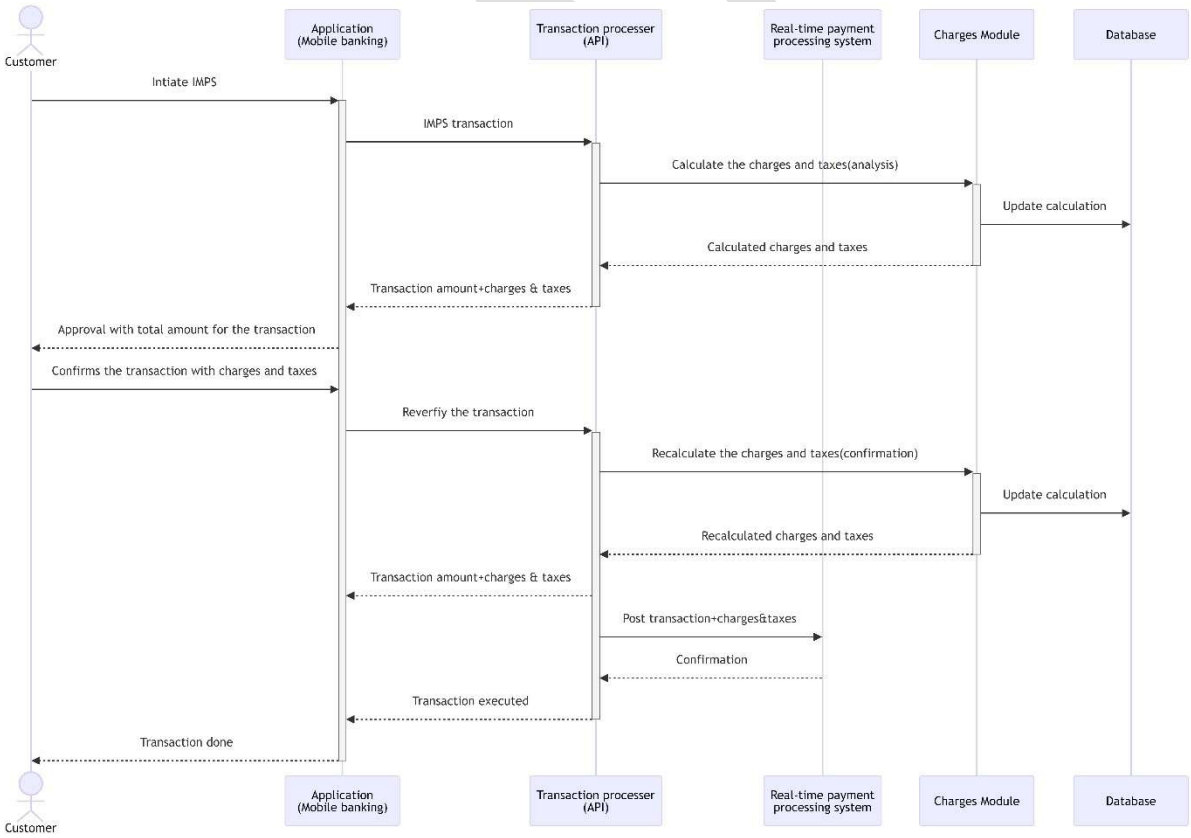
Calculation Logic:

The calculation logic embedded in the charge code design revolves around these attributes and flags. Transaction types drive the initial selection of the applicable charge code, while the effective date ensures temporal accuracy in charge application. The override flag, is activated, allows for real-time adjustments of charges, catering to unique scenarios. The waivable flag enables the system to identify charges eligible for exemptions, contributing to a customer-centric approach. Simultaneously, the tax flag automates tax calculations, promoting accuracy and compliance with regulatory requirements.

Working of Charges Module for a transaction



The above flow diagram shows the basic working of the charges module for a transaction,

Application: It's the medium for the transaction between the customer and the bank. It can be Internet banking, Mobile banking, UPI, etc.

Transaction processor: Elevate and execute the transaction and post the transaction.

Charges module: Verifies and calculates the charges and taxes for the transaction.

Admin User Interface Features

The admin UI incorporates the following features,

- Charge List
- Add Charge
- Edit Charge
- Delete Charge
- Charge Plan List
- Add Charge Plan

  Add Rules
- Edit Charge Plan

  Add/Edit Rules
- Delete Charge Plan

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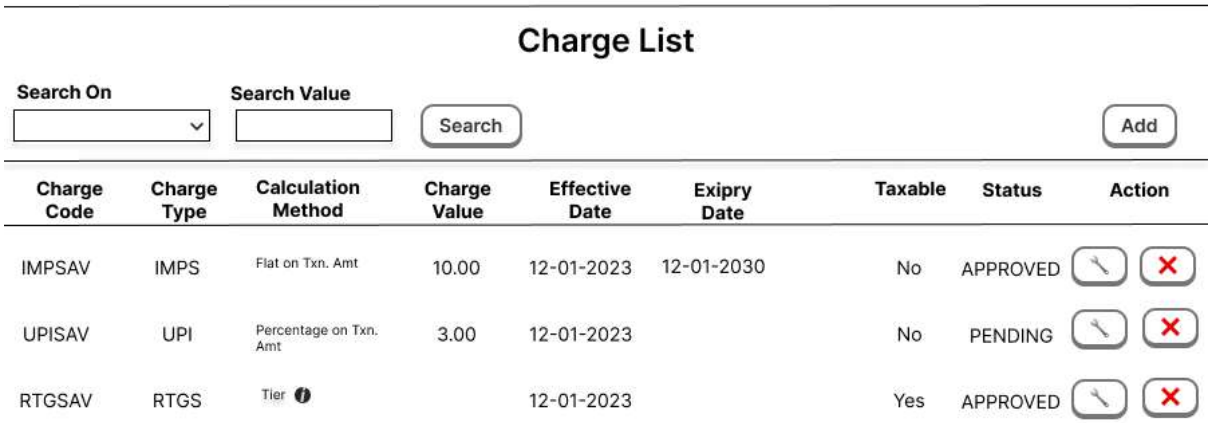
Business Flow

Generating the Charge List



The above wireframe diagram shows the Charge List UI.

The charge list within the charges module is a comprehensive catalog of the charges associated with various banking services.

The charge list typically provides user-friendly interface, allowing the user to easily navigate through the list, understand the details of each charge, and access additional information if needed.

<u>Categorize charges:</u> The list categorizes charges based on transaction types. This categorization simplifies the identification and management of charges.

<u>Charge descriptions:</u> Each charge in the list comes with a detailed description outlining the nature of the charge, the conditions under which it applies, and any relevant parameters that influence the calculation.

<u>Pricing details:</u> The charge list includes pricing details for each charge, specifying whether it's a flat amount or the percentage of the transaction, or follows a tiered charge structure.

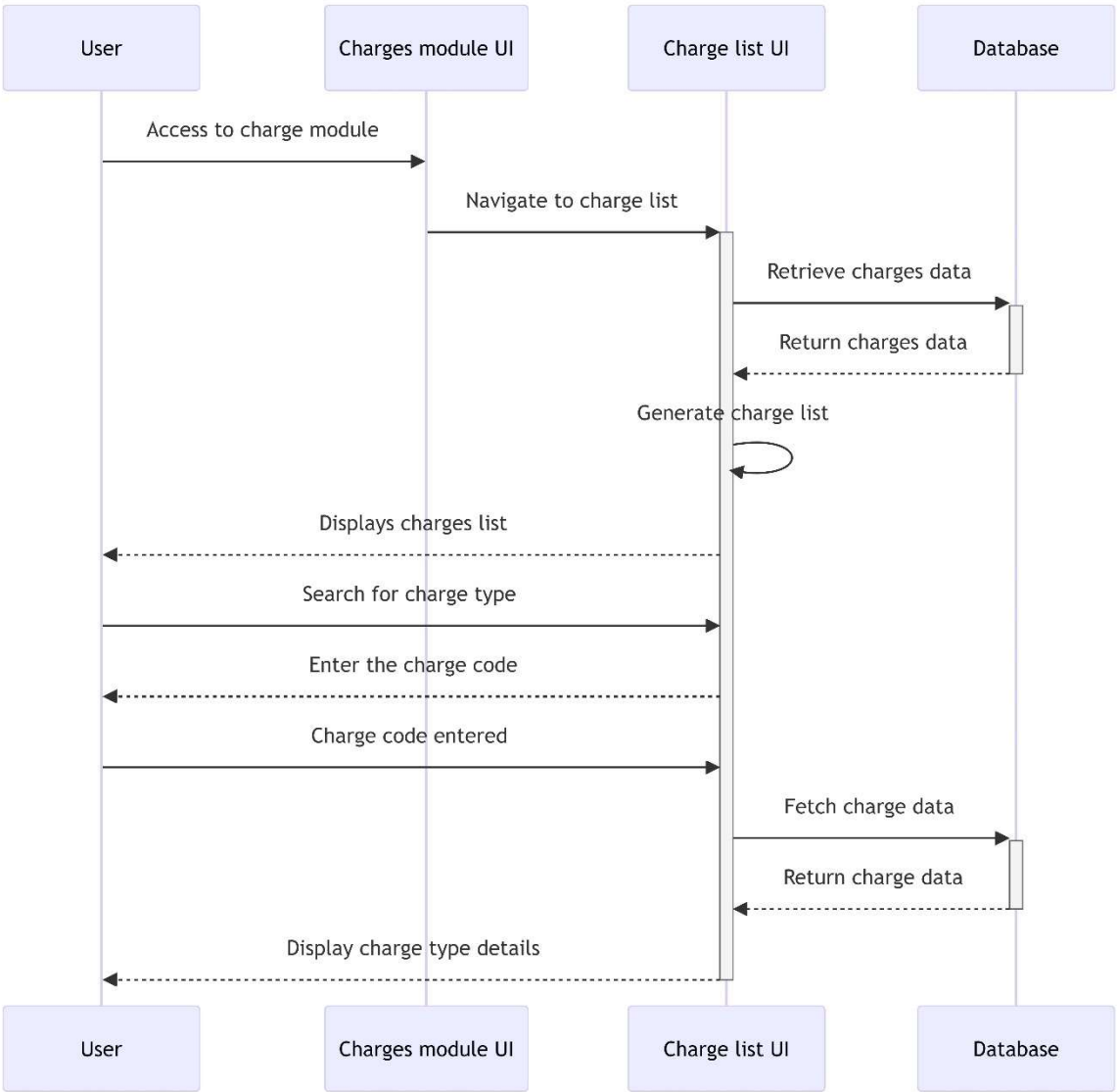
<u>Status indicator:</u> Information about the status of each charge, such as whether the charge is pending approval or approved or rejected, may be displayed.

Search and filter: Utilize search and filter functionalities to locate specific charges quickly, especially in cases where there are numerous charges.

Sortable columns: Columns in the list may be sortable on parameters like charge type, effective date and expiry date.

Action buttons: Common actions, such as editing, deleting, or adding a new charge, are often accessible from the charge list.

The below diagram shows the flow for viewing the charge list and searching the charge type.



Creation of a New Charge type

Add Charge

Code *

Type *

Description

Effective Date *

Expiry Date

Overridable Waivable Tax Applicable

Calculation Method *

Calculation Method Value

(only when Tier method opted)

Seq	Calculation Method *	Value *	Tier Minimum Value *	Tier Maximum Value *
1				
2				

+ Add Tier

Minimum Charge Amount

Maximum Charge Amount

Posting Code

Posting Account

Currency Code

Aggregation Interval *

Add Cancel

The above wireframe diagram shows the Add Charge UI.

<u>Specifying charge type</u>: Determine the charge type of financial transaction for which charges are to be applied, such as transaction charges, ATM withdrawal charges, overdraft charges, etc.

Note- The charges will be configured only for the real-time transaction.

Assigning charges code and description: adding unique codes or identifiers to different transaction types of charges with the description of the charges applied.

Validity period for the applied charges:

- Effective date: The effective date is the point in time when the particular charges come into effect. It marks the beginning of the period during which the specified charge will be applied to the relevant transactions.
- Expiry date: The expiry date represents the end of the validity period for the specific charges. Beyond this date, the charges are no longer applicable. It serves as a cutoff point, indicating when the charge ceases to be enforced.

Note- If the expiry date is not mentioned the charges will be applicable from the effective date mentioned and charges will be valid until the new update of the charge list is configured.

Enabling various flags for a charge: (overridable, waivable, tax applicable)

Specifying Calculation methods and values:

The calculation method for charges in a bank can vary based on the type of charge.

- Flat charge – A fixed amount applied uniformly for a transaction.

  E.g.- ₹5 transaction charge on every IMPS/NEFT transaction.
- Percentage-based charge – A specified percentage of the transaction amount.

  E.g.- 0.5% charge on every Internet banking transaction.
- Tiered-based charge structure – The charge charged depending on the frequency of the transaction.

  The different rates will be predefined in the form of sequences (slabs).

  E.g.- tier 1(sequence 1) - ₹10 charge for the first five ATM transactions per month.

  tier 2(sequence 2) - ₹15 charge for the next five transactions.

  tier 3(sequence 3) - ₹25 for the subsequent withdrawal.
- Aggregation charge structure – The aggregation charge calculation method aggregates or combines multiple small transactions made by a customer within a certain interval period into a single batch transaction and calculates the charge for it. The aggregation charge can be calculated either flat-based or percentage-based.

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If the calculation method is specified as the tiered-based method

- The tier calculation method whether it is flat-based or percentage based and value must be specified for each tier. With the minimum and maximum limit for each tier.

  E.g.-

SLNO		Calculation method	Calculation value	Min Limit	Max Limit
1	Tier 1	Percentage based	0.5%	0	₹10000
2	Tier 2	Percentage Based	0.10%	₹10001	₹100000
3	Tier 3	Percentage based	0.50%	₹100001	Above

Minimum and Maximum charge amount: The minimum and maximum charges are referred to the lowest and highest charges that can be applied for a transaction.

For example,

- The charge configured 0.1% of the transaction amount, to be collected real-time, the transaction amount is ₹10,000/-. The minimum charge amount is specified ₹20/-, then the actual charges are ₹10/- and since the minimum amount is defined as ₹20/-, the final amount collected would be ₹20/-.
- Similarly, if the transaction amount is ₹1,00,000/- and the maximum charge amount is specified as ₹80/-, then the actual charges to be collected is ₹100/- but the maximum charge amount is defined as ₹80/-, the final amount collected would be ₹80/-.
- In the above examples if the minimum and maximum amount is not defined then the actual charges of ₹10/- and ₹100/- would be applied.

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Posting code and posting account: Refers to specific codes and accounts associated with the system to process and record the transaction.

Posting code: This is a code used to categorize and classify transactions for accounting. It helps in organizing transactions based on their such as deposits or withdrawals.

Posting account: This is the account to which a transaction is credited or debited during the posting process.

E.g. – When the deposit is made, the amount is credited to your account, and the bank uses specific accounts for recording the transaction.

Currency code: By default, it is INR.

Aggregation interval: The aggregation interval comes into play, when the aggregation calculation method is used. It can be specified as daily, weekly, or quarterly.

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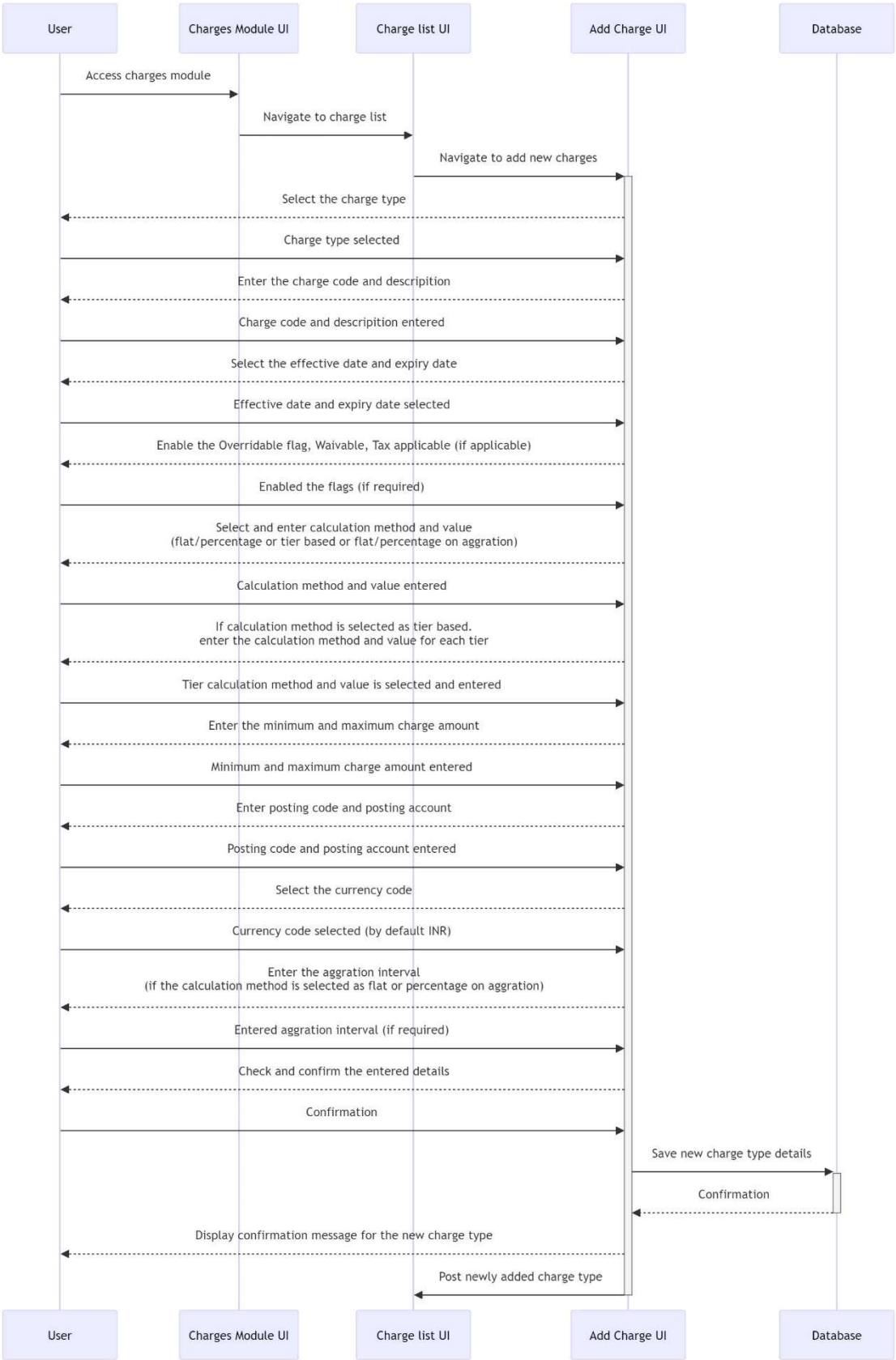
The below diagram shows the flow to add a new charge type.



Access charges module: The user initiates the process by accessing the charges module through the admin application.

Select add charge: Within the charges module, select add charge to add a new charge type.

Enter the charge details: Input the details of the new charge type, including the calculation method and value, minimum and maximum charge amount, currency code, and other relevant information.

Validation and confirm: Perform a validation check on the entered information and confirm to save the new charge type.

Update in database: If the confirmation is received, the system updates the charge details in the database.

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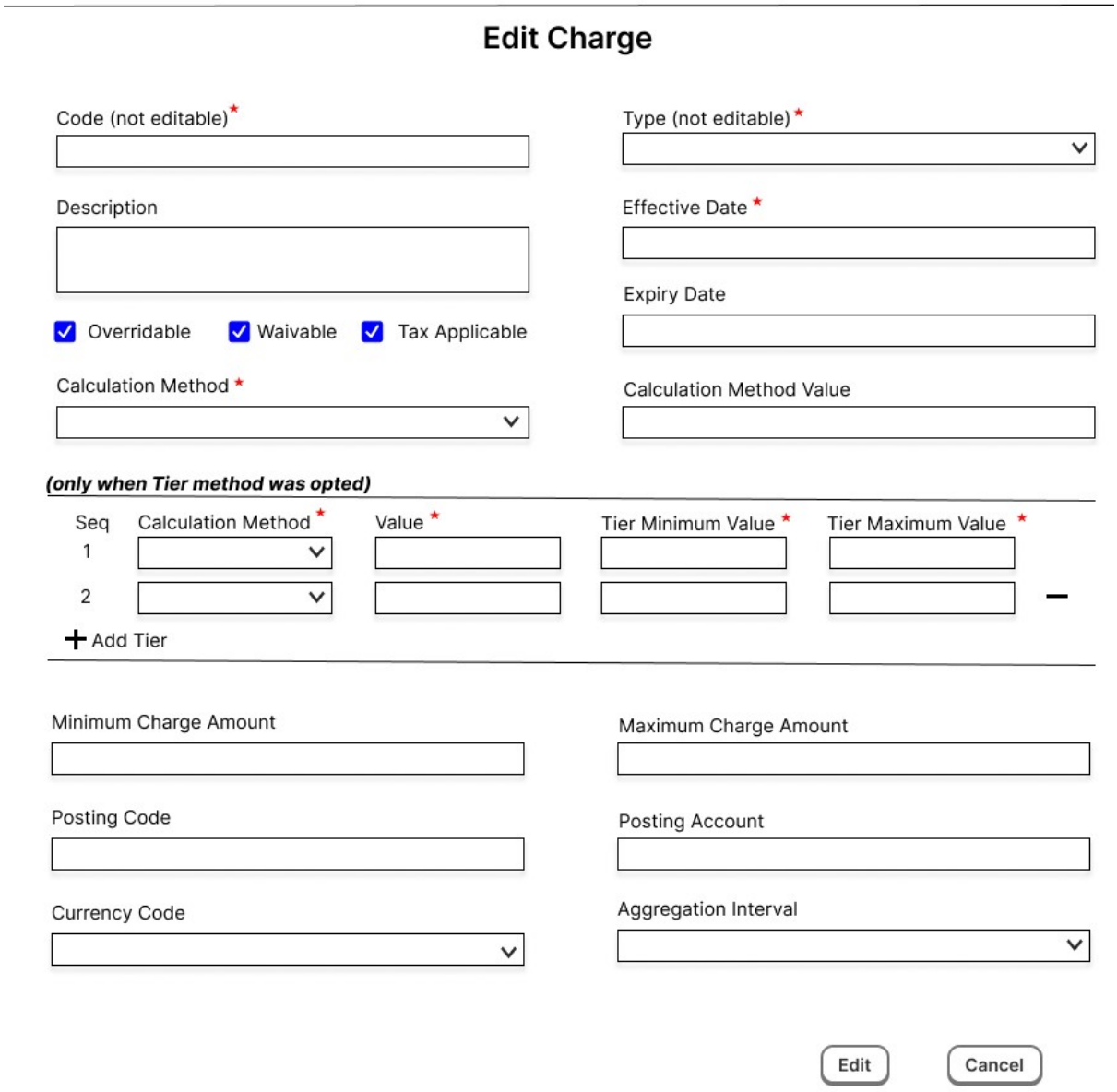
Editing the Existing Charge type



The above wireframe diagram shows the Edit Charge UI.

In the dynamic landscape of banking, situations may arise where adjustments to charges are necessary. The capability to edit charges for the existing transactions allows to maintain accuracy, fairness, and flexibility in the financial operation.

User authorization: Editing charges typically require authorization from bank staff with the appropriate level of access. This ensures that only authorized personnel can modify charges, maintaining security and control.

Search and selection: The system provides a mechanism for searching and selecting specific transaction charges for which charges need to be edited.

Adjustment parameters: A user with editing privileges can modify various parameters associated with charges, including the calculation method and the value, enabling the various flags (overridable, waivable, or tax applicable flags), the minimum and maximum charge amount, posting code and posting account, and currency code. The charge type and charge code cannot be edited.

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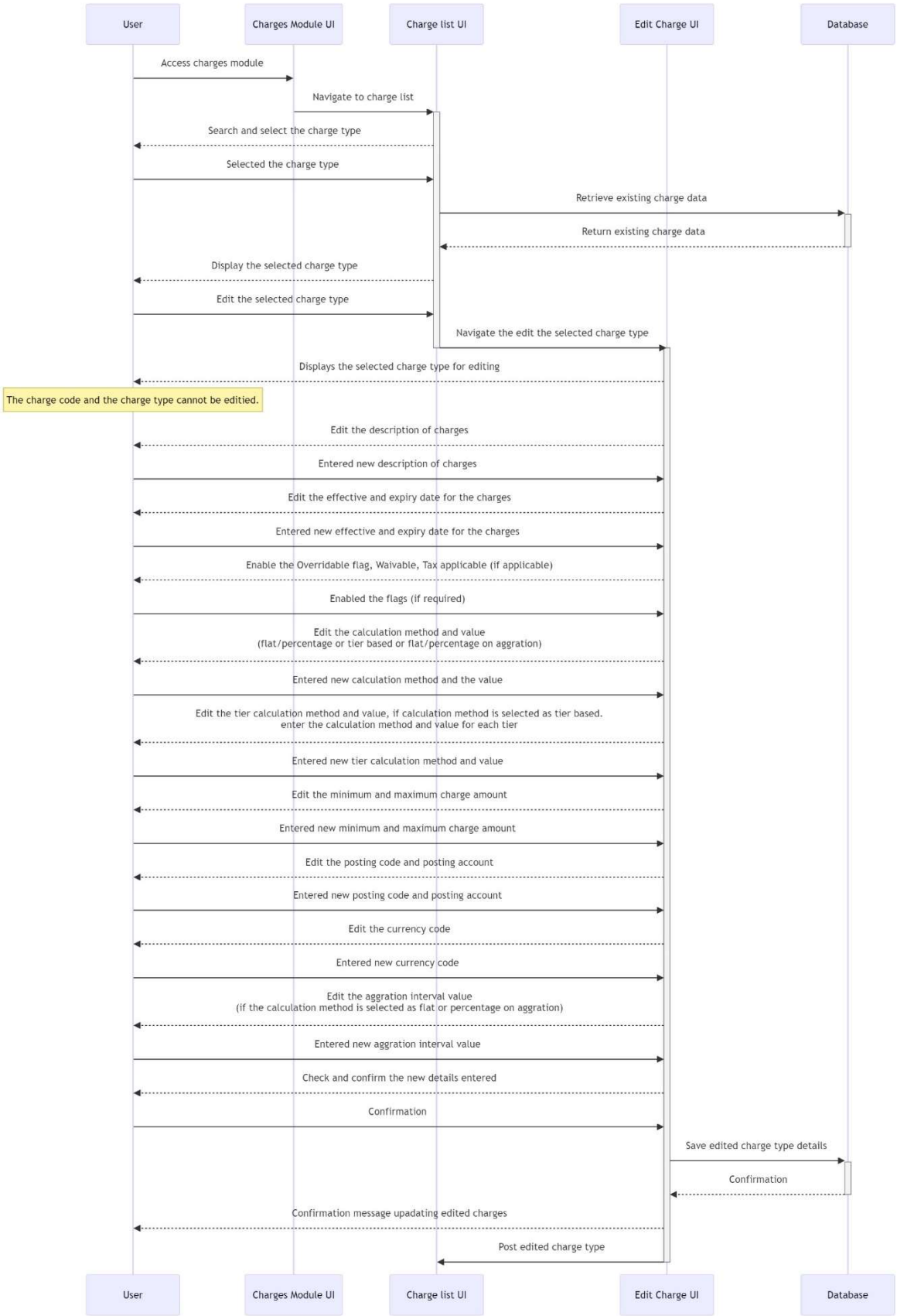
The below diagram shows the flow to edit the existing charge type.



Access charges module: The user initiates the process by accessing the charges module through the admin application.

Search for existing charges: Identify and locate the specific charge to be edited within the charges module. By entering the charge code and charge type.

Select edit option: Choose the edit option associated with the selected charge to initiate the editing process.

Retrieve charge details: The system retrieves and displays the existing details of the selected charge type from the database.

Modify charge details: The User modifies the relevant details of the selected charge and their parameters.

Validation checks: Perform validation checks on the edited information to ensure it aligns with predefined criteria and rules.

Confirmation: The system provides a confirmation prompt to ensure that the user intends to proceed with charges.

Update in database: If the confirmation is received, the system updates the charge details in the database.

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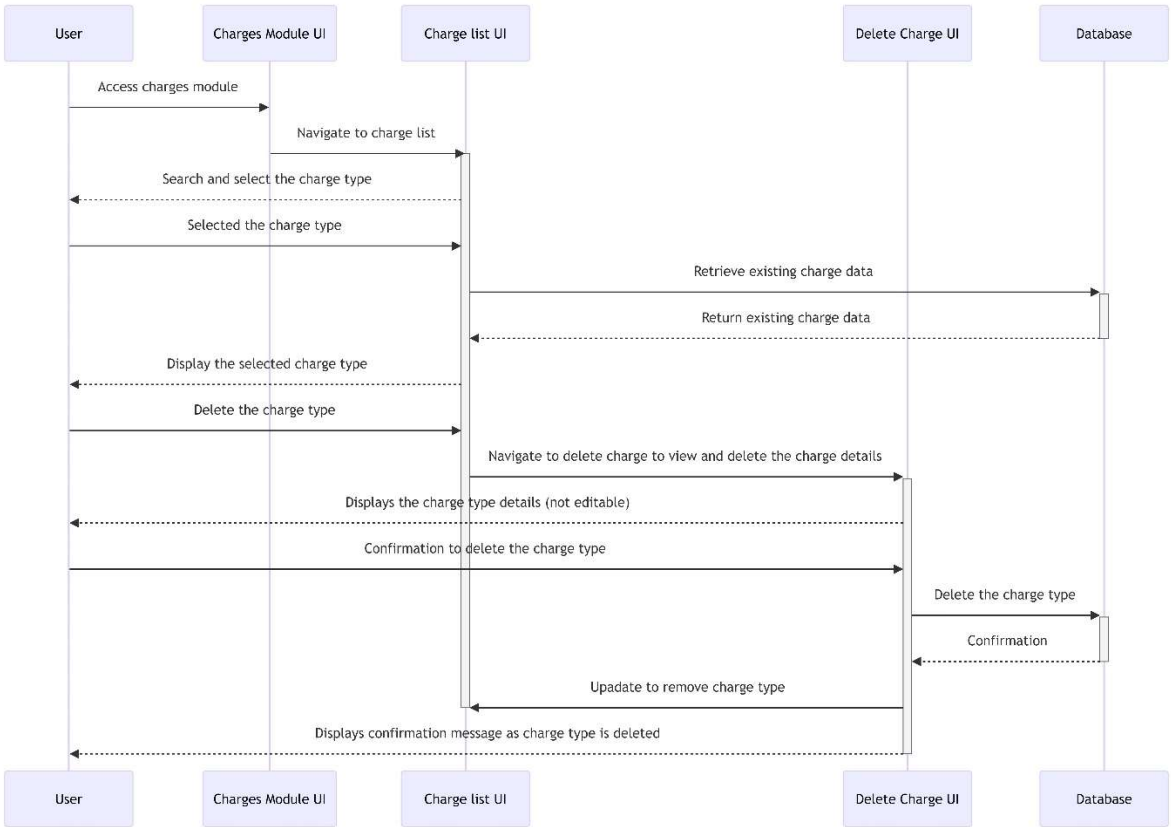
Deleting of Existing Charge type



The above diagram shows the flow for deleting the existing charge type.

<u>Access charges module:</u> The user initiates the process by accessing the charges module through the admin application.

<u>Search for existing charges:</u> Identify and locate the specific charge to be deleted within the charges module. By entering the charge code and charge type.

<u>Select delete option:</u> Choose the delete option associated with the selected charge type to delete.

<u>Retrieve charge details:</u> The system retrieves the charge details of the selected charge type from the database. To prevent accidental deletions, the charge details will be displayed by the system for the user to preview.

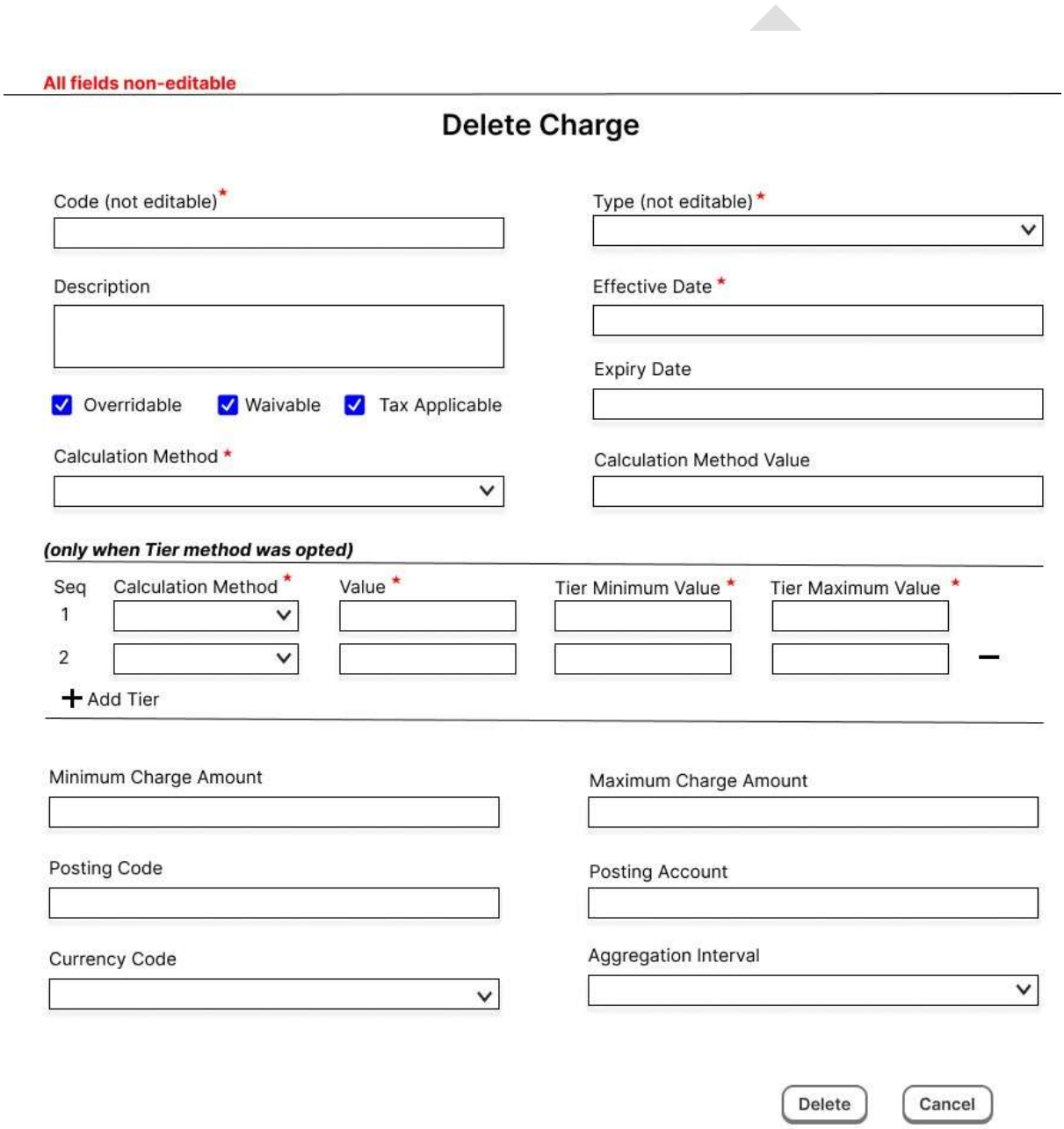
<u>Check for dependencies:</u> Before proceeding with deletion, ensure that the charge type is not being used by any existing charge plans.

<u>Confirm Deletion:</u> After validating details and checking dependencies, confirm to delete the charge.

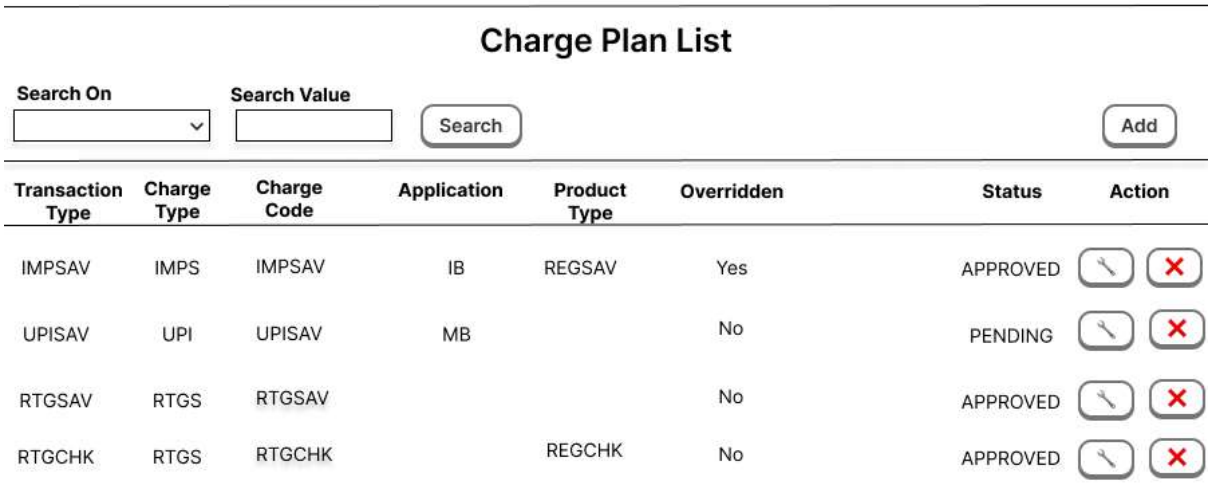
<u>Update in database:</u> If the confirmation is received, the system updates to delete the charge type in the database.

<u>Update in charge list:</u> The system removes the charge type from the charge list.

The below wireframe shows the wireframe of Delete Charge UI.



Generating the Charge Plan List



The above wireframe diagram shows Charge Plan List UI.

<u>Charge plan overview:</u> Explore the charge plan list, which is often presented as a comprehensive view of available plans.

<u>Categorization:</u> Plans may be categorized based on transaction type, product type, and application for easier identification.

<u>Plan details:</u> Each charge plan in the list includes details such as associated charges, terms and conditions, and any specific features.

<u>Edit and manage:</u> The option to edit, manage, or view details of each charge plan are typically available directly from the list.

<u>Status indicators:</u> Information about the status of each charge plan, such as whether the charge plan is pending approval or approved or rejected, may be displayed.

<u>Search and filter:</u> Utilize search and filter functionalities to locate specific charge plans quickly, especially in cases where there are numerous plans.

<u>Sortable columns:</u> Columns in the list may be sortable on parameters like transaction type, product type, and application.

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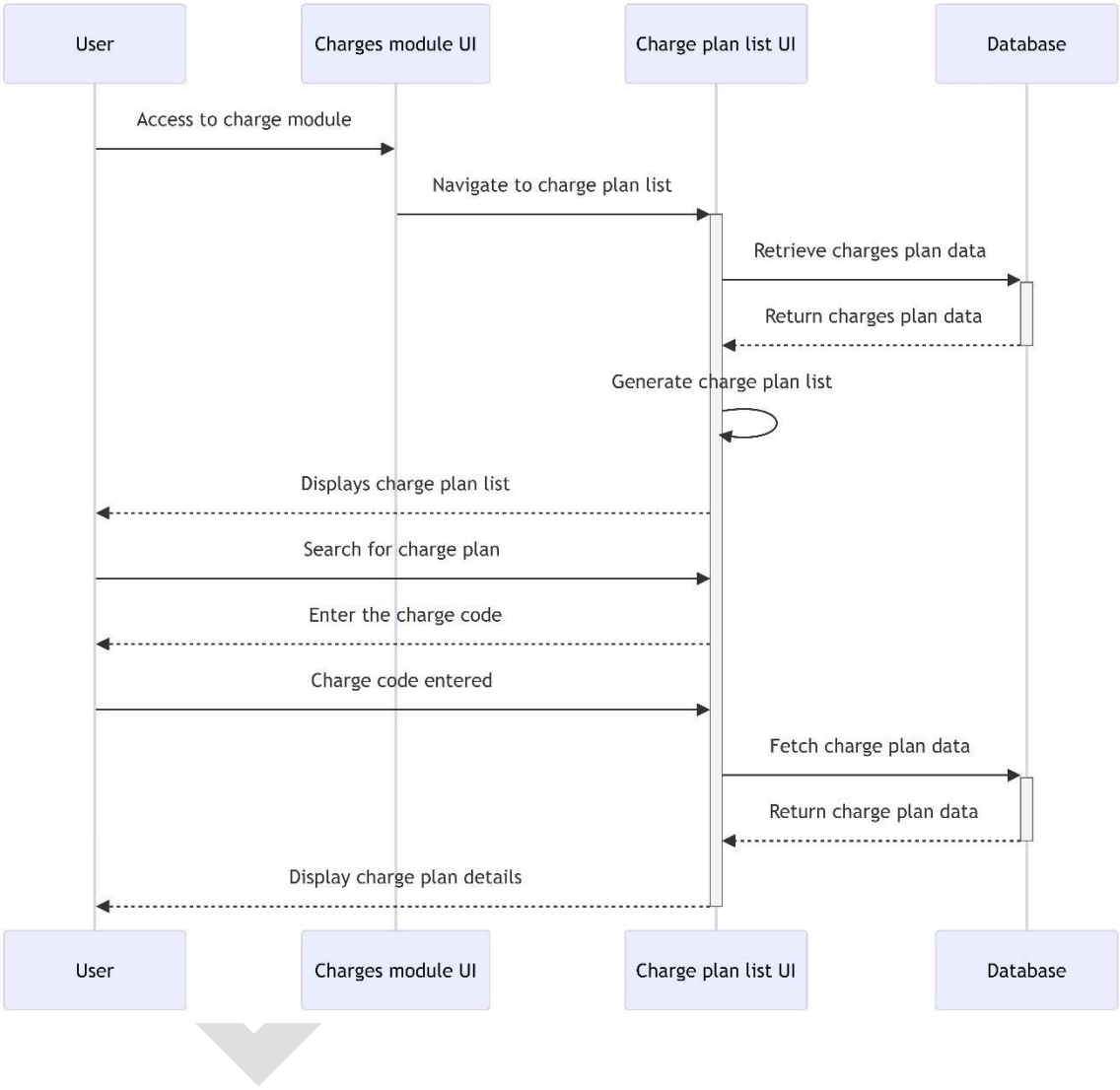
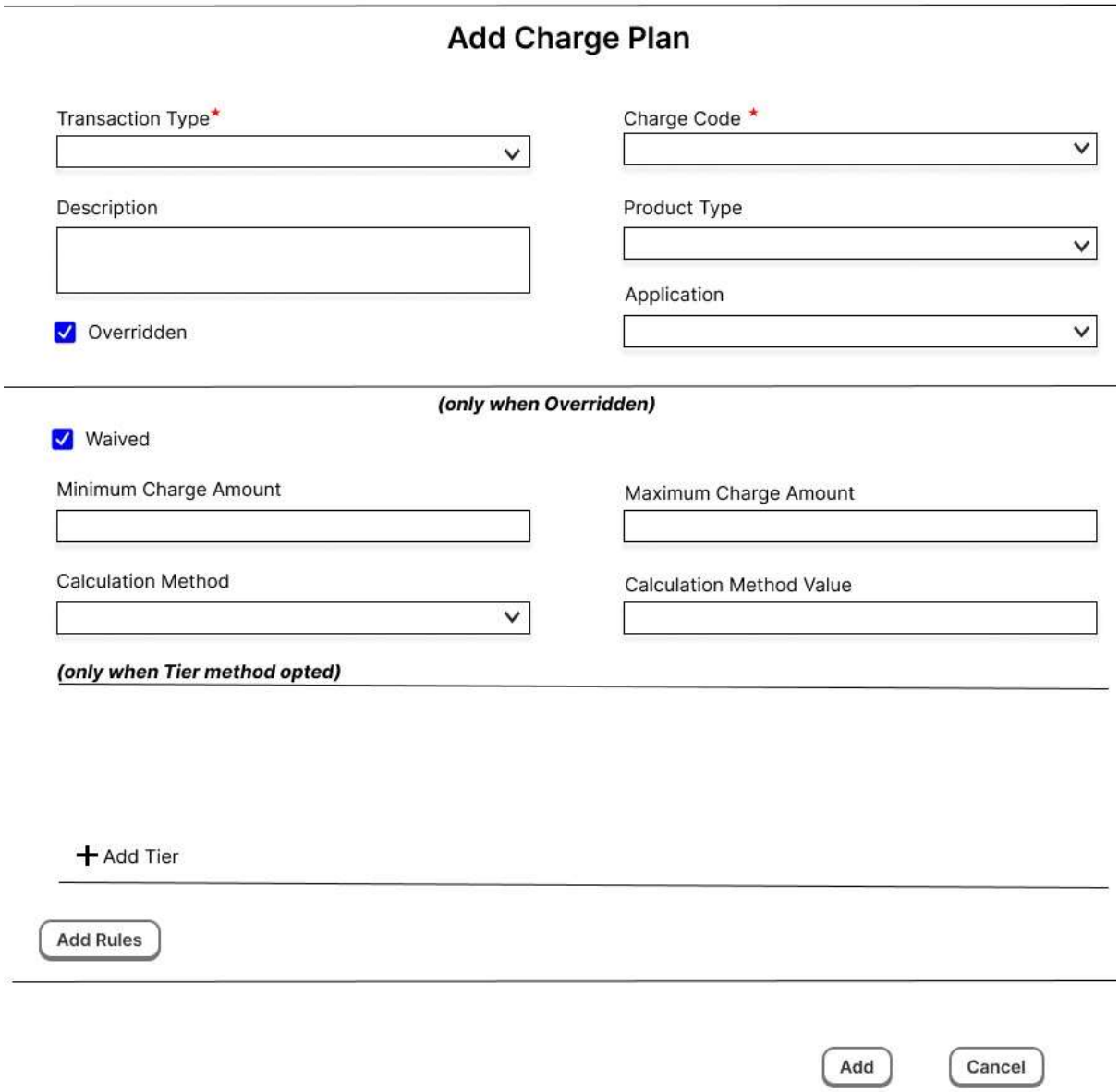
Action buttons: Common actions, such as editing, deleting, or adding a new charge plan, are often accessible from the charge plan list.

The below diagram shows the flow for viewing the charge plan list and searching the charge plan.



Creation of a New Charge Plan type



The above wireframe diagram shows Add Charge Plan UI.

The charge plan within the charges module offers flexibility to adapt its charge structure to changing market conditions, compliance requirements, or internal policies. This adaptability ensures that the bank remains competitive, compliant, and able to meet the diverse needs of its customer base.

Specifying transaction type: Determine the specific type of financial transaction for which charges are to be applied, such as cash withdrawal, IMPS, NEFT, ATM withdrawal, or by an automated process like scheduled payments.

Specifying charges code: Enter the unique codes or identifiers of the specific charge type to be applied for the particular transaction.

Add description: Define the attributes of the charge plan.

Specifying the product type: Determine the product type, such as savings account, current account, overdraft account, fixed deposit account, etc.

Specifying the application type: Determine the application type through which the transaction is done, such as mobile banking, internet banking, UPI, etc.

Enabling overridable flag: The overridable charge system enables banks to dynamically adjust transaction charges. This flexibility goes beyond a fixed charge structure, allowing for real-time modification based on changing factors. Also, the charges can be overridden based on specific conditions. This could include factors such as customer loyalty, transaction volume, or other criteria defined by the bank.

The separate calculation method and the value to be added (whether it's flat/percentage based on transaction amount or tier charge-based). If the calculation method is specified as tier-based method, the tier calculation method must be defined whether it is flat-based or percentage-based and value must be specified for each tier. With the minimum and maximum limit for each tier.

Enabling waivable flag: The charge is subject to being waived based on specific conditions set by the bank. These conditions could include factors, such as customer loyalty, account type, transaction volume, or participation in promotional offers. If the customer satisfies these conditions, no charge will charged for any of the transactions.

Minimum and maximum charge amount: The minimum and maximum charges refer to the lowest and highest charges that can be applied for a transaction.

Add rules: Adding rules in the charges module involves specifying conditions and actions that determine when certain charge plans are applied to customers. These rules can include factors like transaction type, product type, or specific customer attributes.

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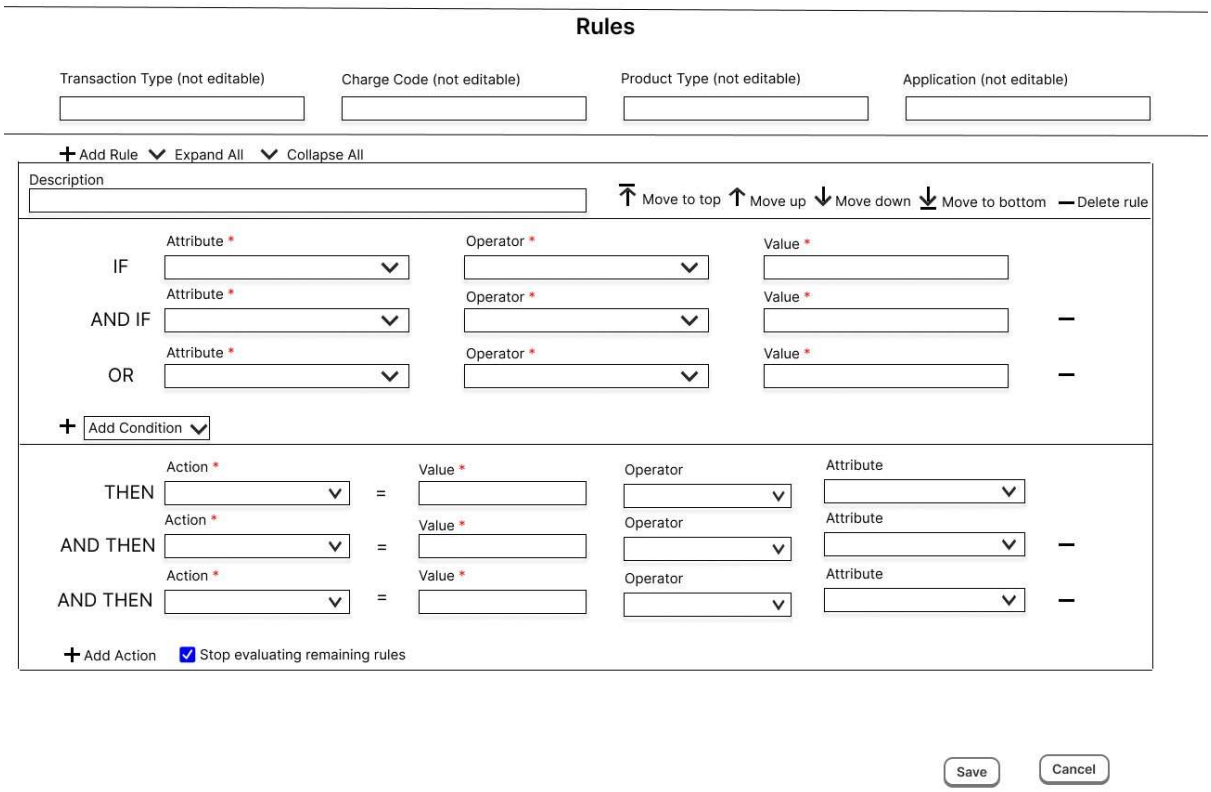
The below wireframe shows the Add Rules UI.



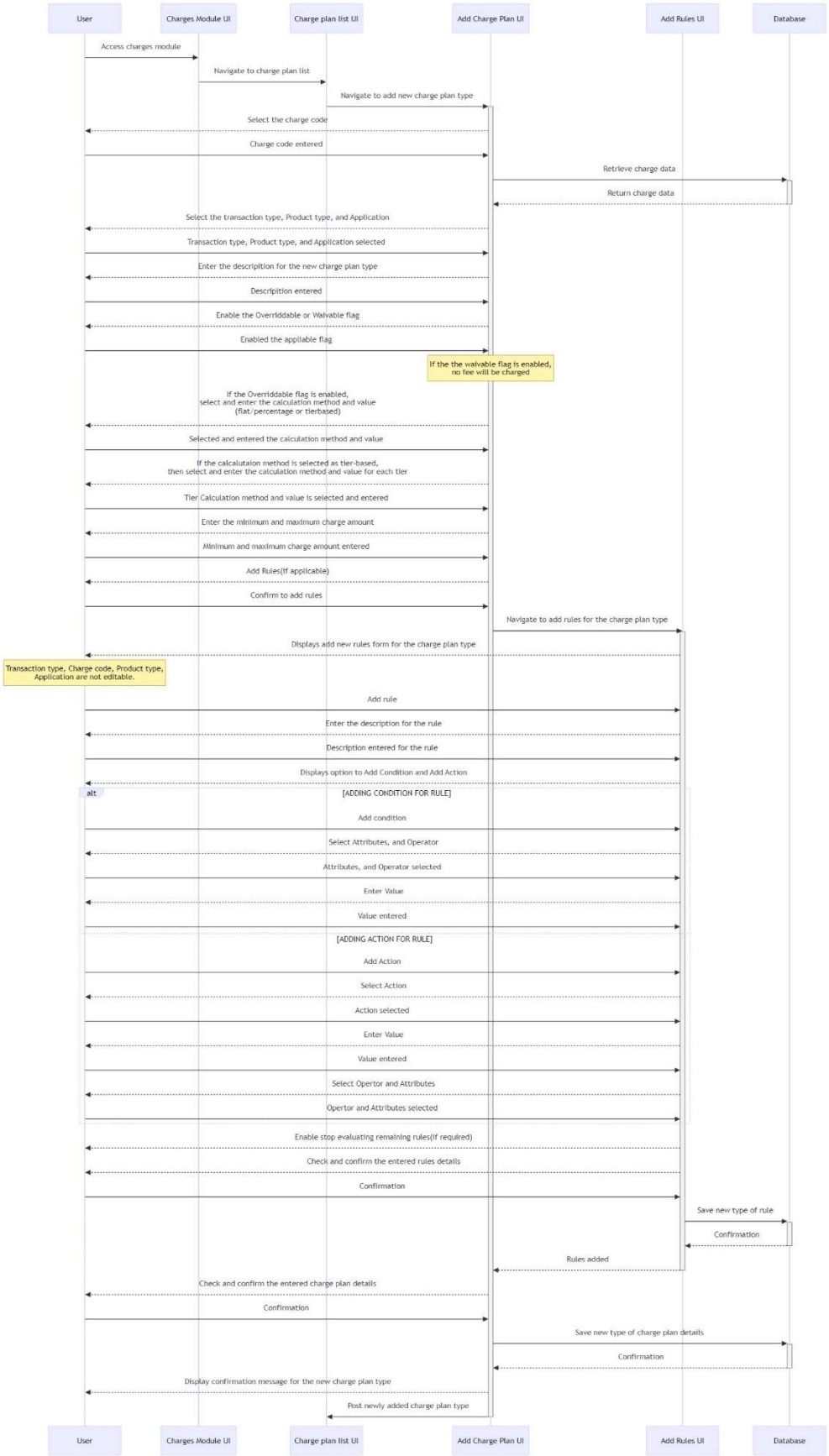
Note: The Transaction type, Charge code, Product type and Application are not editable.

Action buttons: Common actions, such as adding new rules and deleting existing rules, are often accessible with the move up and down of page.

Define Rule Condition: Determine the conditions under which the charge plan type should be applied. These conditions include factors such as customer category, account type, frequency of transaction, and other relevant criteria.

Specify Rule Action: Decide what action should be taken when the rule conditions are met. This could involve fixed charges, calculating a percentage-based charge, waiving the charge, or applying the override for the charge.

The below diagram shows the flow to add a new charge plan type.



Access charges module: The user initiates the process by accessing the charges module through the admin application.

Create a new charge plan: Within the charges module, select add charge plan.

Enter charge plan details: Enter the details of the new charge plan, including transaction type, product type, application, enabling override or waivable flag, minimum and maximum charge amount, adding rules, and other relevant parameters.

Validation and confirm: Perform a validation check on the entered information and confirm to save the new charge plan type.

Update in database: If the confirmation is received, the system updates the charge details in the database.

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Editing the Existing Charge Plan type

Edit Charge Plan

Transaction Type (not editable)*

Charge Code (not editable) *

Description

Product Type

Application

Overridden

(only when Overridden)

Waived

Minimum Charge Amount

Maximum Charge Amount

Calculation Method

Calculation Method Value

(only when Tier method opted)

+ Add Tier

Add/Edit Rules

Edit

Cancel

The above wireframe diagram shows Edit Charge Plan UI.

The Edit Charge Plan within the charges module allows user to modify existing charge structure. This feature provides flexibility to adapt the charge structures in response to changing market conditions, regulatory requirements, or internal policies.

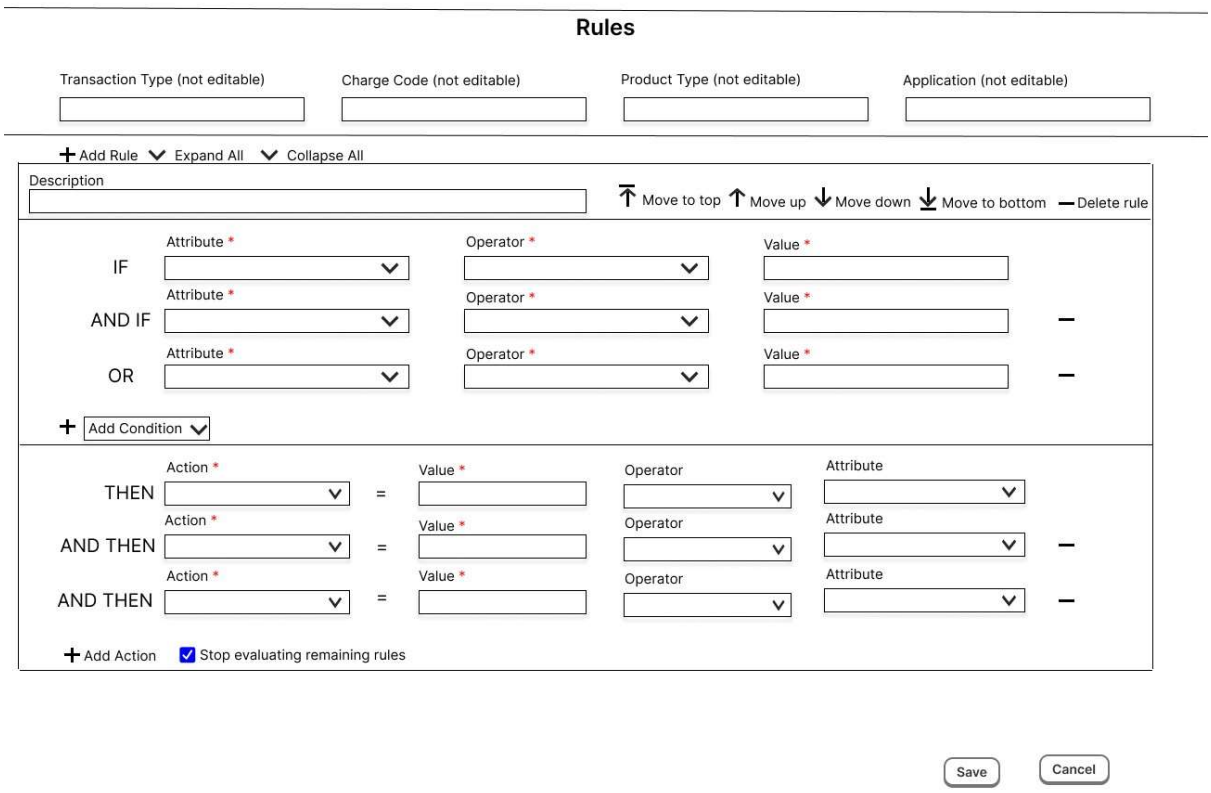
User authorization: Editing charge plan typically require authorization from bank staff with the appropriate level of access. This ensures that only authorized personnel can modify charge plan, maintaining security and control.

Search and selection: The system provides a mechanism for searching and selecting specific type of charge plan for which need to be edited.

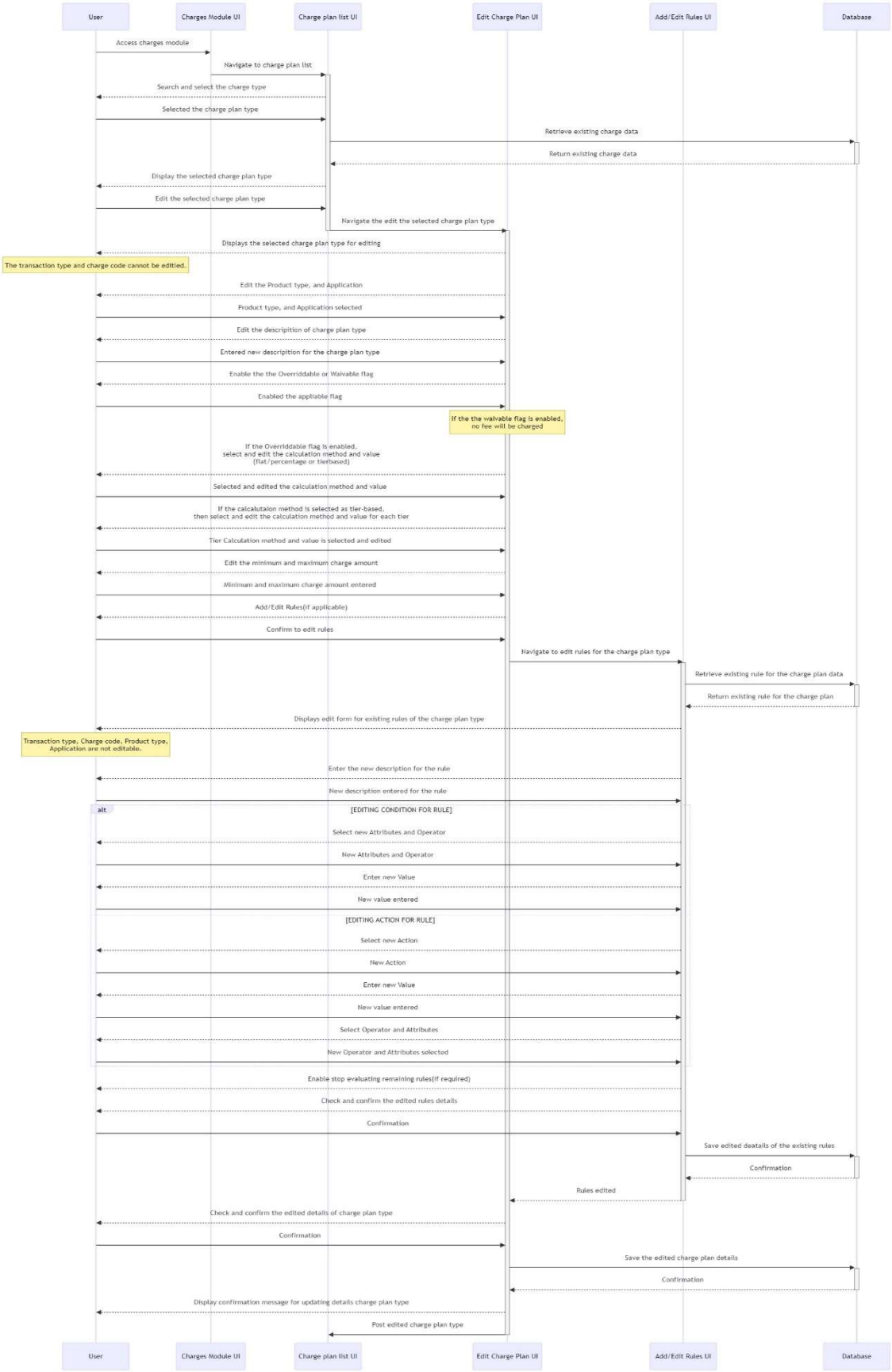
Adjustment parameters: A user with editing privileges can modify various parameters associated with the charge plan, such as product type, application, enabling the override or waivable flag, minimum and maximum charge amount, and Add/Edit Rules. The transaction type and charge code cannot be edited.

Add/edit rules: Adding a rule creates a conditions and actions for applying charges, while editing allows modification to existing conditions and actions.

The below wireframe shows the Add/Edit Rules UI.



The below diagram shows the flow to edit existing charge plan type.



Access to charges module: The user initiates the process by accessing the charges module through the admin application.

Search for existing charge plan: Identify and locate the specific charge to be edited within the charges module. By entering the transaction type and charge code.

Select edit option: Choose the edit option associated with the selected charge to initiate the editing process.

Retrieve charge details: The system retrieves and displays the existing details of the selected charge type from the database.

Modify plan details: The User modifies the relevant details of the selected charge plan and their parameters.

Validation checks: Review the modifications to ensure accuracy and compliance with regulatory standards and internal policies.

Confirmation: The system provides a confirmation prompt to ensure that the user intends to proceed with charges.

Update in database: If the confirmation is received, the system updates the charge details in the database.

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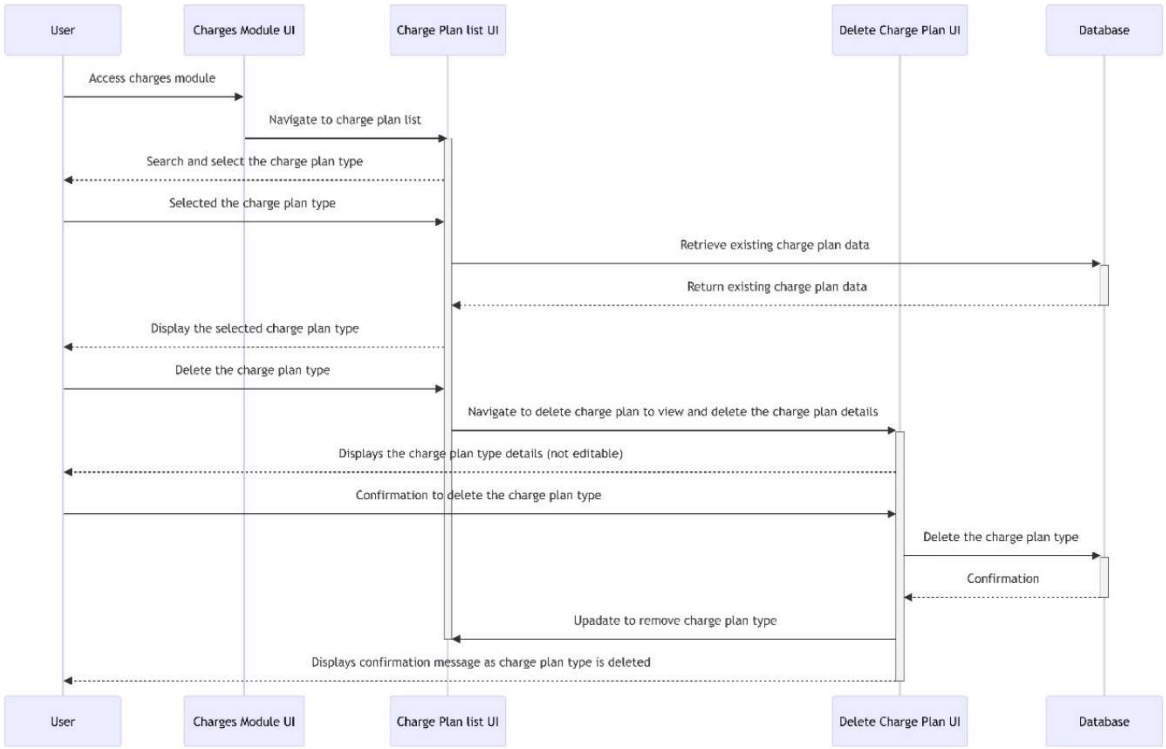
Deleting of Existing Charge plan type



The above diagram shows the flow for deleting the existing charge plan type.

<u>Access charges module:</u> The user initiates the process by accessing the charges module through the admin application.

<u>Search for existing charge plan:</u> Identify and locate the specific charge plan to be deleted within the charges module. By entering the charge code and transaction type.

<u>Select delete option:</u> Choose the delete option associated with the selected charge type to delete.

<u>Retrieve charge details:</u> The system retrieves the charge plan details of the selected charge plan type from the database. To prevent accidental deletions, the charge plan details will be displayed by the system for the user to preview.

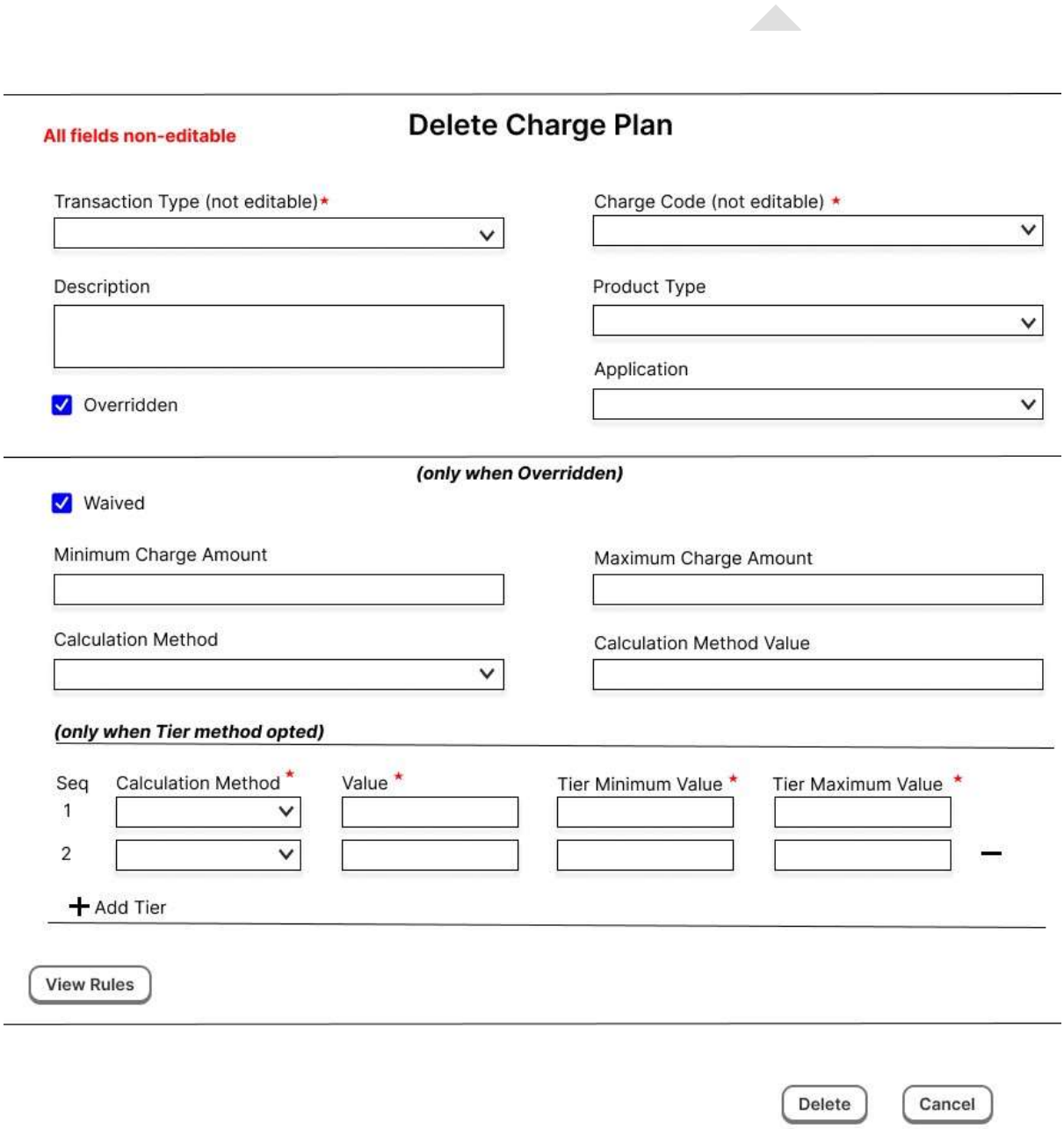
<u>Check for dependencies:</u> Before proceeding with deletion, ensure that the charge plan type is not associated with any accounts.

<u>Confirm Deletion:</u> After validating details and checking dependencies, confirm to delete the charge plan.

<u>Update in database:</u> If the confirmation is received, the system updates to delete the charge plan type in the database.

<u>Update in charge list:</u> The system removes the charge plan type from the charge plan list.

The below wireframe shows the wireframe of Delete Charge Plan UI.



In Scope functionality

- Integration support.

Out of Scope functionality

Any charges configuration outside of the Core application is not in scope.

General Constraints, Assumptions, Dependencies, Guidelines

TBD

REQUIREMENTS

External Interfaces Requirements

User Interfaces

Admin UI

Hardware Interfaces

Software Interfaces

Other Requirements

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Analysis Models

List any attached/referenced documentation such as data flow diagrams, class diagrams, state-transition diagrams, entity-relationship diagrams, etc.

ANALYSIS MODELS		
DOCUMENT NAME	**DESCRIPTION**	**LOCATION**

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Issues List

Detail any unresolved issues.

ISSUES LIST		
ISSUE ID	**ISSUE DESCRIPTION**	**STATUS**

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